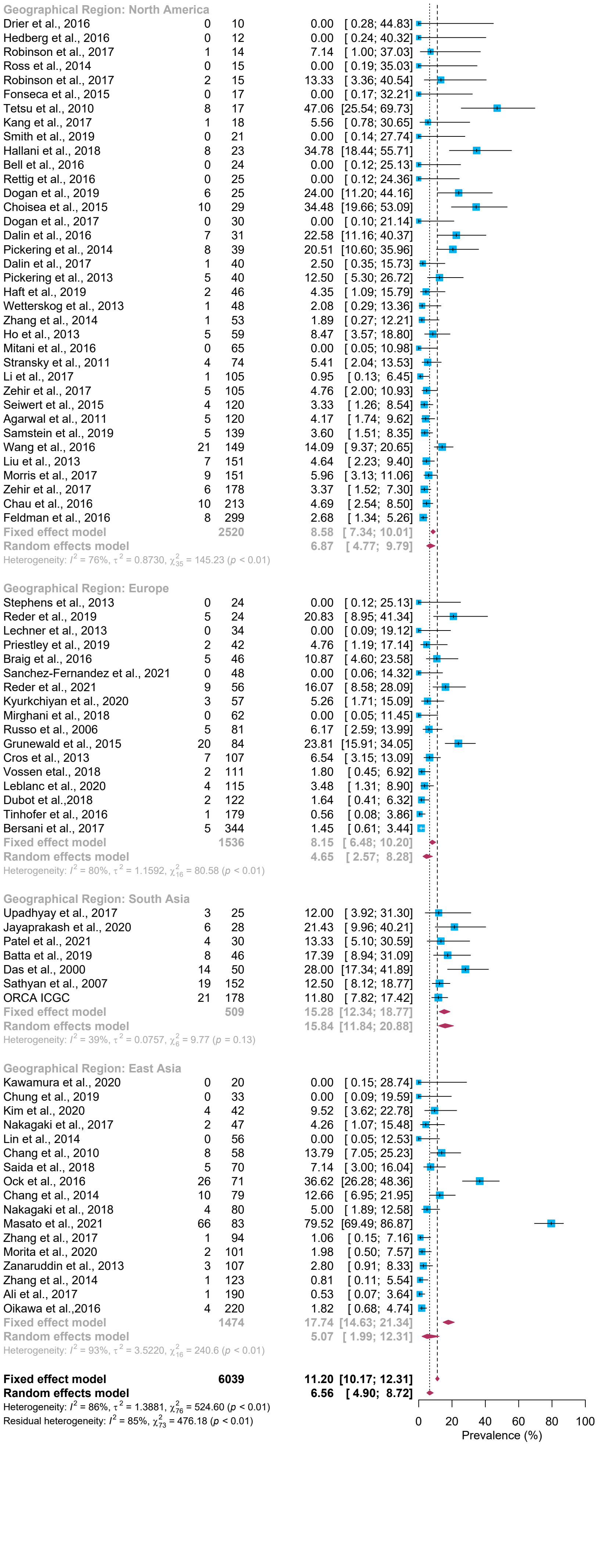
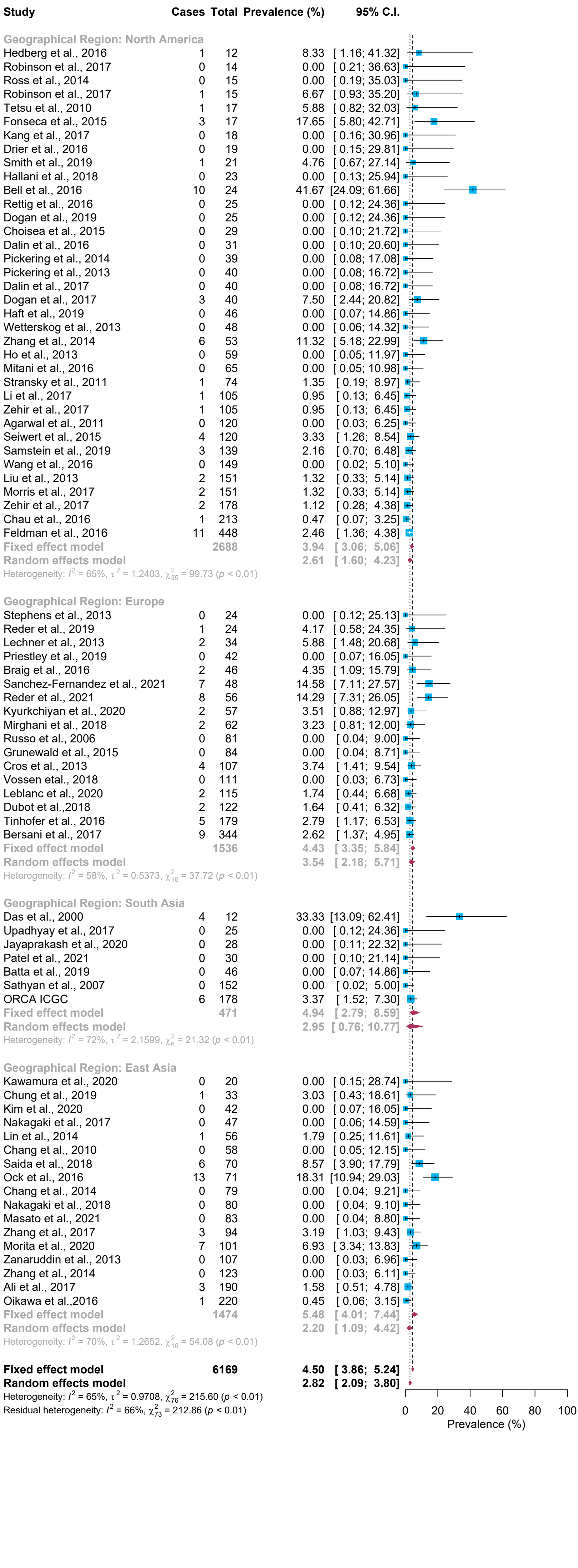
Figure S2: Forest plot of RAS mutation frequency according to geographical region

Forest plot of RAS mutation frequency [%] in head and neck cancer according to geographical region. CI: Confidence interval. I^2: Inconsistency index.

Prevalence of HRAS Mutation



Prevalence of KRAS Mutation



Prevalence of NRAS Mutation

Study	Cases	Total	Prevalence (%)	95% C.I.
Geographical Region: North America				
Drier et al., 2016	0	10	0.00	[0.28; 44.83]
Hedberg et al., 2016	1	12	8.33	[1.16; 41.32]
Robinson et al., 2017 salivary	0	14	0.00	[0.21; 36.63]
Ross et al., 2014	0	15	0.00	[0.19; 35.03]
Robinson et al., 2017	0	15	0.00	[0.19; 35.03]
Tetsu et al., 2010	1	17	5.88	[0.82; 32.03]
Fonseca et al., 2015	1	17	5.88	[0.82; 32.03]
kang et al., 2017	0	18	0.00	[0.16; 30.96]
Smith et al., 2019	0	21	0.00	[0.14; 27.74]
Hallani et al., 2018	0	23	0.00	[0.13; 25.94]
Bell et al., 2016	2	24	8.33	[2.09; 27.88]
Dogan et al., 2019	0	25	0.00	[0.12; 24.36]
Rettig et al., 2016	1	25	4.00	[0.56; 23.55]
Choisea et al., 2015	0	29	0.00	[0.10; 21.72]
Dogan et al., 2017	0	30	0.00	[0.10; 21.14]
Dalin et al., 2016	0	31	0.00	[0.10; 20.60]
Pickering et al., 2014	2	39	5.13	[1.29; 18.32]
Pickering et al., 2013	0	40	0.00	[0.08; 16.72]
Dalin et al., 2017	0	40	0.00	[0.08; 16.72]
Haft et al., 2019	0	46	0.00	[0.07; 14.86]
Wetterskog et al., 2013	0	48	0.00	[0.06; 14.32]
Zhang et al., 2014	6	53	11.32	[5.18; 22.99]
Ho et al., 2013	1	59	1.69	[0.24; 11.07]
Mitani et al., 2016	0	65	0.00	[0.05; 10.98]
Stransky et al., 2011	0	74	0.00	[0.04; 9.77]
Zehir et al., 2017	1	105	0.95	[0.13; 6.45]
Li et al., 2017	3	105	2.86	[0.92; 8.48]
Agarwal et al., 2011	0	120	0.00	[0.03; 6.25]
Seiwert et al., 2015	1	120	0.83	[0.12; 5.67]
Samstein et al., 2019	3	139	2.16	[0.70; 6.48]
Wang et al., 2016	0	149	0.00	[0.02; 5.10]
Liu et al., 2013	0	151	0.00	[0.02; 5.03]
Morris et al., 2017	2	151	1.32	[0.33; 5.14]
Zehir et al., 2017	3	178	1.69	[0.54; 5.09]
Chau et al., 2016	2	213	0.94	[0.23; 3.67]
Feldman et al., 2016	4	381	1.05	[0.39; 2.76]
Fixed effect model		**2602**	**2.45**	**[1.82; 3.28]**
Random effects model			**2.34**	**[1.69; 3.22]**
Heterogeneity: I^2 = 10%, τ^2 = 0.1014, χ^2_{35} = 39.06 (p = 0.29)				
Geographical Region: Europe				
Stephens et al., 2013	0	24	0.00	[0.12; 25.13]
Reder et al., 2019	1	24	4.17	[0.58; 24.35]
Lechner et al., 2013	0	34	0.00	[0.09; 19.12]
Priestley et al., 2019	0	42	0.00	[0.07; 16.05]
Braig et al., 2016	2	46	4.35	[1.09; 15.79]
Sanchez-Fernandez et al., 2021	2	48	4.17	[1.04; 15.19]
Reder et al., 2021	7	56	12.50	[6.08; 23.98]
Kyurkchiyan et al., 2020	1	57	1.75	[0.25; 11.43]
Mirghani et al., 2018	0	62	0.00	[0.05; 11.45]
Russo et al., 2006	0	81	0.00	[0.04; 9.00]
Grunewald et al., 2015	2	84	2.38	[0.60; 9.02]
Cros et al., 2013	2	107	1.87	[0.47; 7.16]
Vossen etal., 2018	1	111	0.90	[0.13; 6.11]
Leblanc et al., 2020	0	115	0.00	[0.03; 6.51]
Dubot et al.,2018	1	122	0.82	[0.12; 5.58]
Tinhofer et al., 2016	2	179	1.12	[0.28; 4.36]
Bersani et al., 2017	4	344	1.16	[0.44; 3.06]
Fixed effect model		**1536**	**2.72**	**[1.88; 3.92]**
Random effects model			**2.08**	**[1.20; 3.60]**
Heterogeneity: I^2 = 46%, τ^2 = 0.5642, χ^2_{16} = 29.42 (p = 0.02)				
Geographical Region: South Asia				
Upadhyay et al., 2017	0	25	0.00	[0.12; 24.36]
Jayaprakash et al., 2020	0	28	0.00	[0.11; 22.32]
Patel et al., 2021	0	30	0.00	[0.10; 21.14]
Batta et al., 2019	0	46	0.00	[0.07; 14.86]
Das et al., 2000	0	50	0.00	[0.06; 13.83]
Sathyan et al., 2007	0	152	0.00	[0.02; 5.00]
ORCA ICGC	2	178	1.12	[0.28; 4.38]
Fixed effect model		**509**	**1.11**	**[0.46; 2.64]**
Random effects model			**1.11**	**[0.46; 2.64]**
Heterogeneity: I^2 = 0%, τ^2 = 0, χ^2_{6} = 1.08 (p = 0.98)				
Geographical Region: East Asia				
Kawamura et al., 2020	0	20	0.00	[0.15; 28.74]
Chung et al., 2019	0	33	0.00	[0.09; 19.59]
Kim et al., 2020	0	42	0.00	[0.07; 16.05]
Nakagaki et al., 2017	0	47	0.00	[0.06; 14.59]
Lin et al., 2014	1	56	1.79	[0.25; 11.61]
Chang et al., 2010	0	58	0.00	[0.05; 12.15]
Saida et al., 2018	0	70	0.00	[0.04; 10.27]
Chang et al., 2014	0	79	0.00	[0.04; 9.21]
Nakagaki et al., 2018	0	80	0.00	[0.04; 9.10]
Masato et al., 2021	0	83	0.00	[0.04; 8.80]
Zhang et al., 2017	1	94	1.06	[0.15; 7.16]
Morita et al., 2020	0	101	0.00	[0.03; 7.35]
Zanaruddin et al., 2013	0	107	0.00	[0.03; 6.96]
Zhang et al., 2014	5	123	4.07	[1.70; 9.39]
Ali et al., 2017	4	190	2.11	[0.79; 5.47]
Oikawa et al.,2016	2	220	0.91	[0.23; 3.56]
Fixed effect model		**1403**	**1.66**	**[1.05; 2.60]**
Random effects model			**1.66**	**[1.05; 2.60]**
Heterogeneity: I^2 = 0%, τ^2 = 0, χ^2_{15} = 9.18 (p = 0.87)				
Fixed effect model		**6050**	**2.24**	**[1.83; 2.74]**
Random effects model			**2.08**	**[1.66; 2.59]**
Heterogeneity: I^2 = 11%, τ^2 = 0.1046, χ^2_{75} = 84.37 (p = 0.21)				
Residual heterogeneity: I^2 = 9%, χ^2_{72} = 78.75 (p = 0.27)				

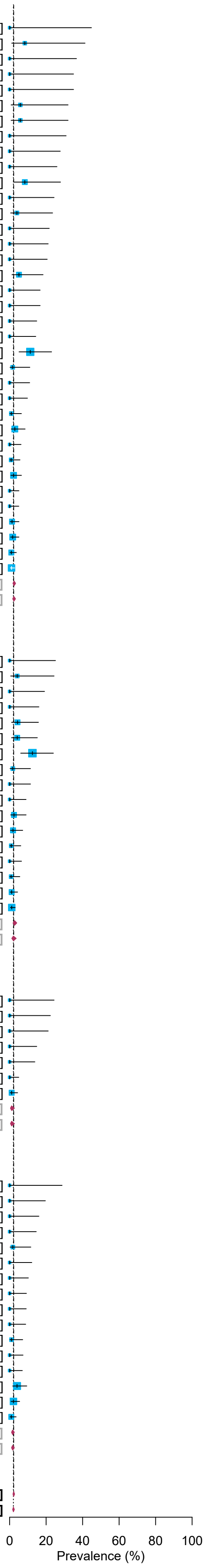
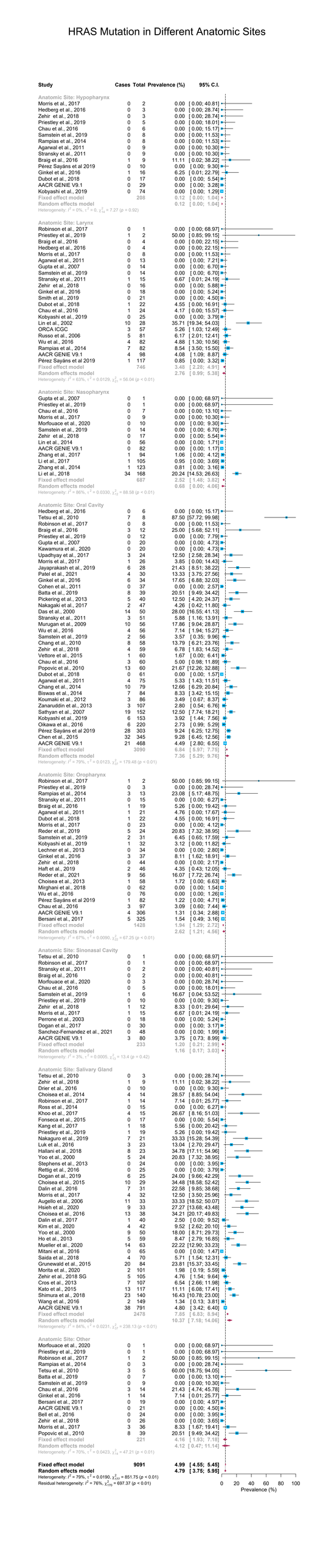


Figure S3: Mutation prevalence according to anatomical site

Forest plot of RAS mutation frequency [%] in head and neck cancer according to tumor anatomical site of origin. CI: Confidence interval. I^2: Inconsistency index.

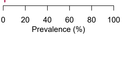
HRAS Mutation in Different Anatomic Sites



KRAS Mutation in Different Anatomic Sites

Study	Cases	Total	Prevalence (%)	95% C.I.
Anatomic Site: Hypopharynx				
Fury et al., 2013	0	1	0.00	[0.00; 68.97]
Prigge et al., 2014	0	1	0.00	[0.00; 68.97]
Carvalho et al., 2013	0	2	0.00	[0.00; 40.81]
Morris et al., 2017	0	2	0.00	[0.00; 40.81]
Westbrook et al., 2019	0	2	0.00	[0.00; 40.81]
Hedberg et al., 2016	0	3	0.00	[0.00; 28.74]
Zehir et al., 2018	0	3	0.00	[0.00; 28.74]
Priestley et al., 2019	0	5	0.00	[0.00; 18.01]
Chau et al., 2016	0	6	0.00	[0.00; 15.17]
Gauthaman et al., 2020	4	6	66.67	[27.79; 95.43]
Samstein et al., 2019	0	8	0.00	[0.00; 11.53]
Agarwal et al., 2011	0	9	0.00	[0.00; 10.30]
Braig et al., 2016	1	9	11.11	[0.02; 38.22]
Stransky et al., 2011	1	9	11.11	[0.02; 38.22]
Pérez Sayáns et al 2019	0	10	0.00	[0.00; 9.30]
Bissada et al., 2013	0	11	0.00	[0.00; 8.48]
Sheikh Ali et al., 2008	0	14	0.00	[0.00; 6.70]
Dubot et al., 2018	0	17	0.00	[0.00; 5.54]
Weber et al., 2003	2	18	11.11	[1.18; 29.19]
Suda et al., 2012	0	25	0.00	[0.00; 3.79]
AACR GENIE V9.1	0	29	0.00	[0.00; 3.28]
Fuji et al., 2013	0	83	0.00	[0.00; 1.15]
Fixed effect model		273	0.43	[0.05; 1.58]
Random effects model			0.84	[0.00; 3.18]
Heterogeneity: I^2 = 36%, τ^2 = 0.0135, χ^2_{21} = 33.8 (p = 0.04)				
Anatomic Site: Larynx				
Robinson et al., 2017	0	1	0.00	[0.00; 68.97]
Priestley et al., 2019	0	2	0.00	[0.00; 40.81]
Braig et al., 2016	1	4	25.00	[0.11; 72.04]
Hedberg et al., 2016	1	4	25.00	[0.11; 72.04]
Westbrook et al., 2019	1	7	14.29	[0.03; 47.26]
Morris et al., 2017	0	8	0.00	[0.00; 11.53]
Agarwal et al., 2011	0	13	0.00	[0.00; 7.21]
Carvalho et al., 2013	0	14	0.00	[0.00; 6.70]
Gupta et al., 2007	0	14	0.00	[0.00; 6.70]
Samstein et al., 2019	1	14	7.14	[0.01; 25.77]
Stransky et al., 2011	0	15	0.00	[0.00; 6.27]
Zehir et al., 2018	1	16	6.25	[0.01; 22.79]
Sheikh Ali et al., 2008	0	17	0.00	[0.00; 5.54]
Ruiz-Godoy et al., 2006	0	20	0.00	[0.00; 4.73]
Su et al., 2017	1	21	4.76	[0.00; 17.67]
Dubot et al., 2018	0	22	0.00	[0.00; 4.30]
Suda et al., 2012	0	23	0.00	[0.00; 4.12]
Chau et al., 2016	0	24	0.00	[0.00; 3.95]
Weber et al., 2003	1	25	4.00	[0.00; 14.98]
Bissada et al., 2013	3	29	10.34	[2.10; 23.78]
Fuji et al., 2013	0	51	0.00	[0.00; 1.87]
Kyurkchiyan et al., 2020	2	57	3.51	[0.34; 9.79]
Russo et al., 2006	1	81	0.00	[0.00; 1.18]
Wu et al., 2016	9	82	10.98	[5.17; 18.60]
AACR GENIE V9.1	4	98	4.08	[1.09; 8.87]
Pérez Sayáns et al 2019	0	117	0.00	[0.00; 0.82]
Fixed effect model		779	1.40	[0.69; 2.34]
Random effects model			1.43	[0.34; 3.25]
Heterogeneity: I^2 = 58%, τ^2 = 0.0121, χ^2_{25} = 59.74 (p < 0.01)				
Anatomic Site: Nasopharynx				
Fury et al., 2013	0	1	0.00	[0.00; 68.97]
Gupta et al., 2007	0	1	0.00	[0.00; 68.97]
Priestley et al., 2019	0	1	0.00	[0.00; 68.97]
Westbrook et al., 2019	0	1	0.00	[0.00; 68.97]
Chau et al., 2016	0	7	0.00	[0.00; 13.10]
Morris et al., 2017	0	9	0.00	[0.00; 10.30]
Morfouace et al., 2020	0	10	0.00	[0.00; 9.30]
Samstein et al., 2019	0	14	0.00	[0.00; 6.70]
Zehir et al., 2018	0	17	0.00	[0.00; 5.54]
Chou et al., 2008	1	45	0.00	[0.00; 2.12]
Lin et al., 2014	1	56	1.79	[0.00; 6.86]
Lin et al., 2014	1	66	1.52	[0.00; 5.84]
Zhang et al., 2015	1	70	1.43	[0.00; 5.51]
AACR GENIE V9.1	2	82	2.44	[0.24; 6.86]
Zhang et al., 2017	3	94	3.19	[0.62; 7.67]
Li et al., 2017	1	105	0.95	[0.00; 3.69]
Zhang et al., 2014	0	123	0.00	[0.00; 0.78]
Fixed effect model		702	0.83	[0.29; 1.63]
Random effects model			0.83	[0.29; 1.63]
Heterogeneity: I^2 = 0%, τ^2 = 0, χ^2_{16} = 12.9 (p = 0.68)				
Anatomic Site: Oral Cavity				
Fury et al., 2013	1	1	100.00	[31.03; 100.00]
Prigge et al., 2014	0	3	0.00	[0.00; 28.74]
Hedberg et al., 2016	0	6	0.00	[0.00; 15.17]
Westbrook et al., 2019	0	6	0.00	[0.00; 15.17]
Robinson et al., 2017	0	8	0.00	[0.00; 11.53]
Tetsu et al., 2010	1	8	12.50	[0.02; 42.28]
Braig et al., 2016	0	12	0.00	[0.00; 7.79]
Das et al., 2000	4	12	33.33	[10.66; 61.20]
Priestley et al., 2019	0	12	0.00	[0.00; 7.79]
Weber et al., 2003	1	13	7.69	[0.01; 27.57]
Gupta et al., 2007	0	20	0.00	[0.00; 4.73]
Kawamura et al., 2020	0	20	0.00	[0.00; 4.73]
Bissada et al., 2013	0	21	0.00	[0.00; 4.50]
Upadhyay et al., 2017	0	25	0.00	[0.00; 3.79]
Kucuk et al., 2016	0	26	0.00	[0.00; 3.65]
Morris et al., 2017	1	26	3.85	[0.00; 14.43]
Jayaprakash et al., 2019	0	28	0.00	[0.00; 3.39]
Patel et al., 2021	0	30	0.00	[0.00; 3.17]
Suda et al., 2012	2	31	6.45	[0.65; 17.59]
Cohen et al., 2011	0	37	0.00	[0.00; 2.57]
Batta et al., 2019	0	39	0.00	[0.00; 2.44]
Pickering et al., 2013	0	40	0.00	[0.00; 2.38]
Bruckman et al., 2010	1	42	2.38	[0.00; 9.08]
Gauthaman et al., 2020	25	44	56.82	[42.10; 70.95]
Nakagaki et al., 2017	0	47	0.00	[0.00; 2.03]
Wang et al., 2011	0	47	0.00	[0.00; 2.03]
Sheikh Ali et al., 2008	0	51	0.00	[0.00; 1.87]
Stransky et al., 2011	0	51	0.00	[0.00; 1.87]
Samstein et al., 2019	1	56	1.79	[0.00; 6.86]
Wu et al., 2016	1	56	1.79	[0.00; 6.86]
Chang et al., 2010	0	58	0.00	[0.00; 1.65]
Zehir et al., 2018	1	59	1.69	[0.00; 6.52]
Chau et al., 2016	1	60	1.67	[0.00; 6.41]
Dubot et al., 2018	0	61	0.00	[0.00; 1.57]
Carvalho et al., 2013	0	66	0.00	[0.00; 1.45]
Tan et al., 2014	0	66	0.00	[0.00; 1.45]
Agarwal et al., 2011	0	75	0.00	[0.00; 1.28]
Chang et al., 2014	0	79	0.00	[0.00; 1.21]
Zanaruddin et al., 2013	0	107	0.00	[0.00; 0.89]
Sathyan et al., 2007	0	152	0.00	[0.00; 0.63]
Oikawa et al., 2016	1	220	0.45	[0.00; 1.77]
Pérez Sayáns et al 2019	1	303	0.33	[0.00; 1.29]
Zehir et al., 2018	8	345	2.32	[0.84; 4.17]
AACR GENIE V9.1	8	468	1.71	[0.73; 3.08]
Fixed effect model		2937	0.72	[0.45; 1.06]
Random effects model			0.70	[0.17; 1.59]
Heterogeneity: I^2 = 77%, τ^2 = 0.0129, χ^2_{43} = 184.65 (p < 0.01)				
Anatomic Site: Oropharynx				
Robinson et al., 2017	0	2	0.00	[0.00; 40.81]
Gauthaman et al., 2020	2	3	66.67	[14.42; 99.75]
Priestley et al., 2019	0	3	0.00	[0.00; 28.74]
Westbrook et al., 2019	0	7	0.00	[0.00; 13.10]
Sheikh Ali et al., 2008	0	9	0.00	[0.00; 10.30]
Carvalho et al., 2013	0	12	0.00	[0.00; 7.79]
Fury et al., 2013	0	14	0.00	[0.00; 6.70]
Stransky et al., 2011	0	15	0.00	[0.00; 6.27]
Braig et al., 2016	0	19	0.00	[0.00; 4.97]
Friedland et al., 2011	0	20	0.00	[0.00; 4.73]
Agarwal et al., 2011	0	21	0.00	[0.00; 4.50]
Prigge et al., 2014	0	21	0.00	[0.00; 4.50]
Dubot et al., 2018	2	22	9.09	[0.94; 24.28]
van Damme et al., 2010	1	22	4.55	[0.00; 16.91]
Morris et al., 2017	0	23	0.00	[0.00; 4.12]
Reder et al., 2019	1	24	4.17	[0.00; 15.57]
Suda et al., 2012	1	25	4.00	[0.00; 14.98]
Samstein et al., 2019	1	33	3.03	[0.00; 11.47]
Weber et al., 2003	1	33	3.03	[0.00; 11.47]
Lechner et al., 2013	2	34	5.88	[0.59; 16.11]
Lin et al., 2014	1	37	2.70	[0.00; 10.27]
Haft et al., 2019	0	46	0.00	[0.00; 2.07]
Zehir et al., 2018	0	52	0.00	[0.00; 1.84]
Reder et al., 2021	8	56	14.29	[6.44; 24.56]
Mirghani et al., 2018	2	62	3.23	[0.31; 9.02]
Fuji et al., 2013	0	71	0.00	[0.00; 1.35]
Wu et al., 2016	8	76	10.53	[4.68; 18.36]
Pérez Sayáns et al 2019	0	82	0.00	[0.00; 1.17]
Chau et al., 2016	0	97	0.00	[0.00; 0.99]
Bissada et al., 2013	4	123	3.25	[0.86; 7.10]
AACR GENIE V9.1	4	306	1.31	[0.34; 2.88]
Bersani et al., 2017	9	325	2.77	[1.27; 4.83]
Fixed effect model		1695	1.68	[1.12; 2.34]
Random effects model			1.49	[0.60; 2.77]
Heterogeneity: I^2 = 62%, τ^2 = 0.0081, χ^2_{31} = 80.95 (p < 0.01)				
Anatomic Site: Sinonasal Cavity				
Fury et al., 2013	0	1	0.00	[0.00; 68.97]
Robinson et al., 2017	0	1	0.00	[0.00; 68.97]
Tetsu et al., 2010	0	1	0.00	[0.00; 68.97]
Braig et al., 2016	0	2	0.00	[0.00; 40.81]
Stransky et al., 2011	0	2	0.00	[0.00; 40.81]
Morfouace et al., 2020	0	3	0.00	[0.00; 28.74]
Chau et al., 2016	0	5	0.00	[0.00; 18.01]
Samstein et al., 2019	0	6	0.00	[0.00; 15.17]
Priestley et al., 2019	0	10	0.00	[0.00; 9.30]
Suda et al., 2012	0	11	0.00	[0.00; 8.48]
Zehir et al., 2018	0	12	0.00	[0.00; 7.79]
Morris et al., 2017	1	15	6.67	[0.01; 24.19]
Frattini et al., 2006	9	18	50.00	[27.71; 72.29]
Franchi et al., 2014	1	27	3.70	[0.01; 13.91]
Szablewski et al., 2013	12	28	42.86	[25.43; 61.26]
Dogan et al., 2017	3	30	10.00	[2.03; 23.04]
Projetti et al., 2013	2	34	5.88	[0.59; 16.11]
Lopez et al., 2012 and Garcia-Inclan et al., 2012	7	58	12.07	[5.03; 21.61]
AACR GENIE V9.1	0	80	0.00	[0.00; 1.20]
Wang et al., 2019	30	80	37.50	[27.27; 48.32]
Bornholdt et al., 2008	8	174	4.60	[2.00; 8.20]
Fixed effect model		598	8.06	[6.02; 10.38]
Random effects model			5.67	[1.33; 12.74]
Heterogeneity: I^2 = 85%, τ^2 = 0.0549, χ^2_{25} = 132.66 (p < 0.01)				
Anatomic Site: Salivary Gland				
Tetsu et al., 2010	0	3	0.00	[0.00; 28.74]
Zehir et al., 2018	0	9	0.00	[0.00; 10.30]
Drier et al., 2016	0	10	0.00	[0.00; 9.30]
Robinson et al., 2017	0	14	0.00	[0.00; 6.70]
Khoo et al., 2017	4	15	26.67	[8.16; 51.03]
Ross et al., 2014	0	15	0.00	[0.00; 6.27]
Fonseca et al., 2015	3	17	17.65	[3.79; 38.68]
Fu et al., 2015	2	18	11.11	[1.18; 29.19]
Kang et al., 2017	0	18	0.00	[0.00; 5.24]
Priestley et al., 2019	0	19	0.00	[0.00; 4.97]
Shalmon et al., 2016	0	21	0.00	[0.00; 4.50]
Hallani et al., 2018	0	23	0.00	[0.00; 4.12]
Luk et al., 2016	0	23	0.00	[0.00; 4.12]
Stephens et al., 2013	0	24	0.00	[0.00; 3.95]
Yoo et al., 2000	2	24	8.33	[0.86; 22.95]
Dogan et al., 2019	0	25	0.00	[0.00; 3.79]
Rettig et al., 2016	0	25	0.00	[0.00; 3.79]
Chowalee et al., 2015	0	29	0.00	[0.00; 3.28]
Dalin et al., 2016	0	31	0.00	[0.00; 3.07]
Morris et al., 2017	4	32	12.50	[3.50; 25.96]
Augello et al., 2006	2	33	6.06	[0.61; 16.58]
Dalin et al., 2017	0	40	0.00	[0.00; 2.38]
Kim et al., 2020	0	42	0.00	[0.00; 2.27]
Schneider et al., 2013	0	43	0.00	[0.00; 2.22]
Ho et al., 2013	0	59	0.00	[0.00; 1.62]
Mueller et al., 2020	3	63	4.76	[0.93; 11.34]
Dahse et al., 2009	1	65	1.54	[0.00; 5.93]
Mitani et al., 2016	0	65	0.00	[0.00; 1.47]
Saida et al., 2018	6	70	8.57	[3.20; 16.20]
Grunewald et al., 2015	0	84	0.00	[0.00; 1.14]
Morita et al., 2020	7	101	6.93	[2.83; 12.67]
Zehir et al., 2018 SG	1	105	0.95	[0.00; 3.69]
Cros et al., 2013	4	107	3.74	[0.99; 8.14]
Shimura et al., 2018	0	140	0.00	[0.00; 0.68]
Wang et al., 2016	0	149	0.00	[0.00; 0.64]
AACR GENIE V9.1	8	791	1.01	[0.43; 1.83]
Fixed effect model		2352	0.94	[0.59; 1.37]
Random effects model			0.98	[0.33; 1.96]
Heterogeneity: I^2 = 67%, τ^2 = 0.0086, χ^2_{35} = 104.75 (p < 0.01)				
Anatomic Site: Other				
Morfouace et al., 2020	0	1	0.00	[0.00; 68.97]
Priestley et al., 2019	0	1	0.00	[0.00; 68.97]
Robinson et al., 2017	1	2	50.00	[0.85; 99.15]
Gauthaman et al., 2020	3	4	75.00	[27.96; 99.89]
Tetsu et al., 2010	0	5	0.00	[0.00; 18.01]
Batta et al., 2019	0	7	0.00	[0.00; 13.10]
Samstein et al., 2019	0	9	0.00	[0.00; 10.30]
Bissada et al., 2013	0	11	0.00	[0.00; 8.48]
Chau et al., 2016	0	14	0.00	[0.00; 6.70]
Ginkel et al., 2016	1	14	7.14	[0.01; 25.77]
Zehir et al., 2018	0	18	0.00	[0.00; 5.24]
Bersani et al., 2017	0	19	0.00	[0.00; 4.97]
AACR GENIE V9.1	0	21	0.00	[0.00; 4.50]
Bell et al., 2016	10	24	41.67	[23.12; 61.53]
Morris et al., 2017	3	36	8.33	[1.67; 19.41]
Popovic et al., 2010	0	39	0.00	[0.00; 2.44]
Sanchez-Fernandez et al., 2021	7	48	14.58	[6.14; 25.84]
Fixed effect model		273	4.11	[2.08; 6.78]
Random effects model			3.67	[0.30; 10.55]
Heterogeneity: I^2 = 77%, τ^2 = 0.0552, χ^2_{23} = 70.15 (p < 0.01)				
Fixed effect model		**9609**	**1.31**	**[1.09; 1.54]**
Random effects model			**1.40**	**[0.92; 1.97]**

Heterogeneity: I^2 = 73%, τ^2 = 0.0150, χ^2_{214} = 779.08 (p < 0.01)
Residual heterogeneity: I^2 = 70%, χ^2_{207} = 679.61 (p < 0.01)



NRAS Mutation in Different Anatomic Sites

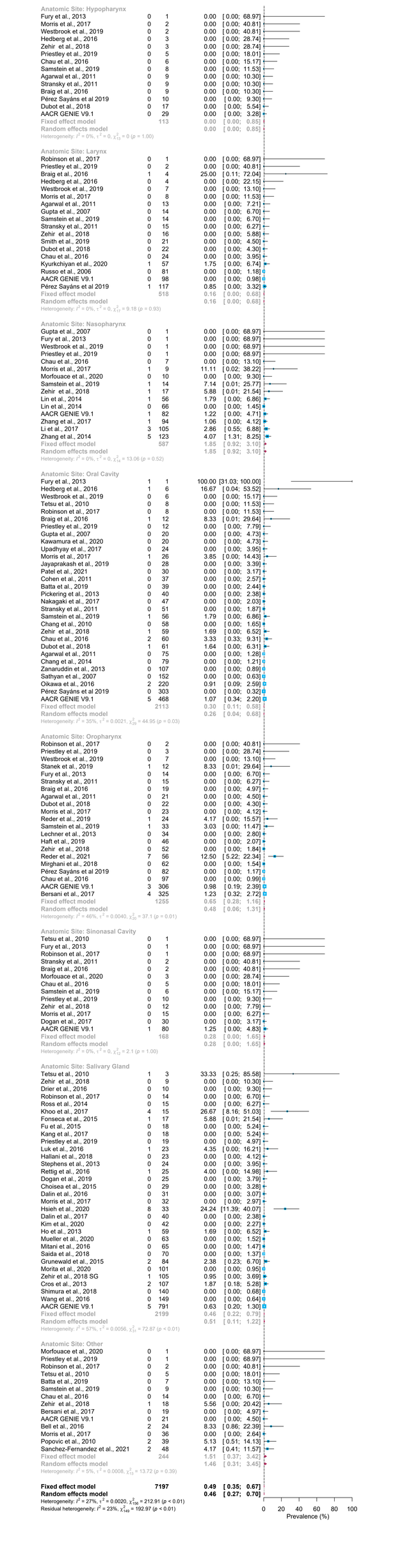
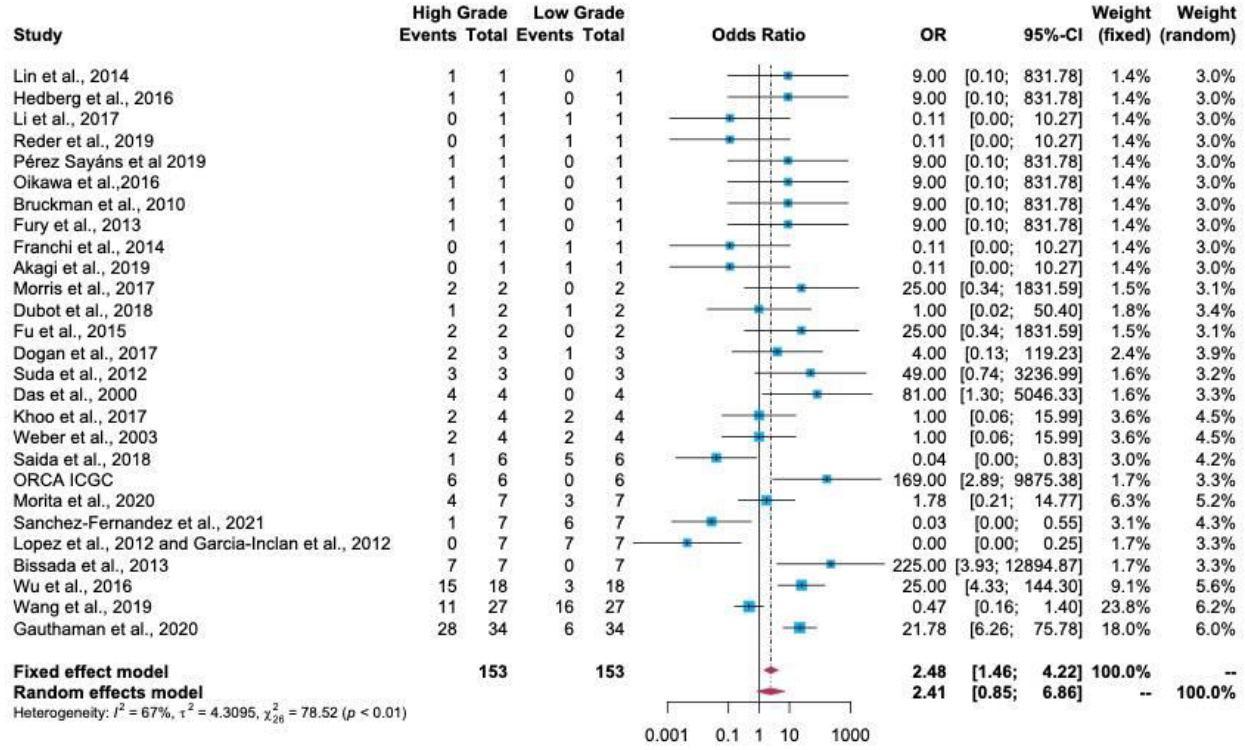


Figure S4: Association between RAS Mutations and Disease Stage/Grade

An odds ratio analysis of the association between tumor grade and KRAS or NRAS mutations; no statistically significant correlations were found.

KRAS



NRAS

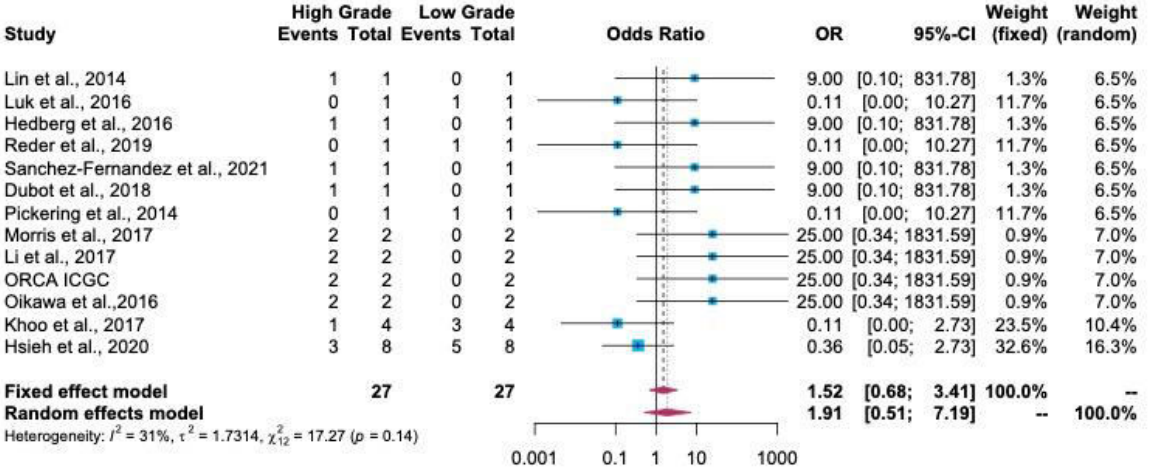
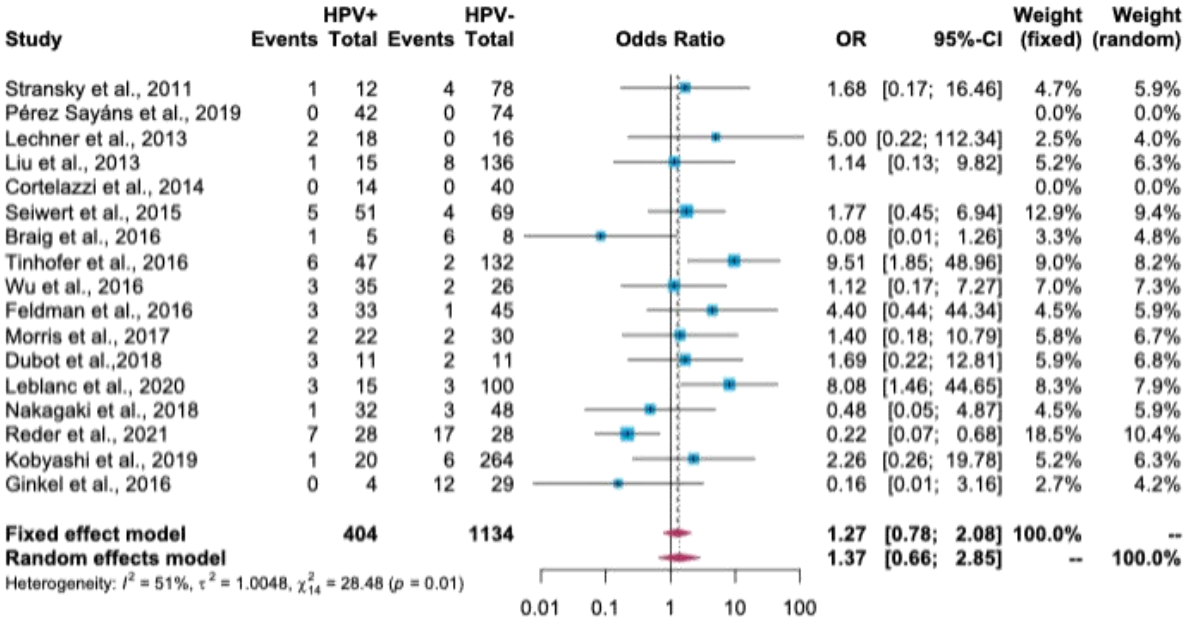
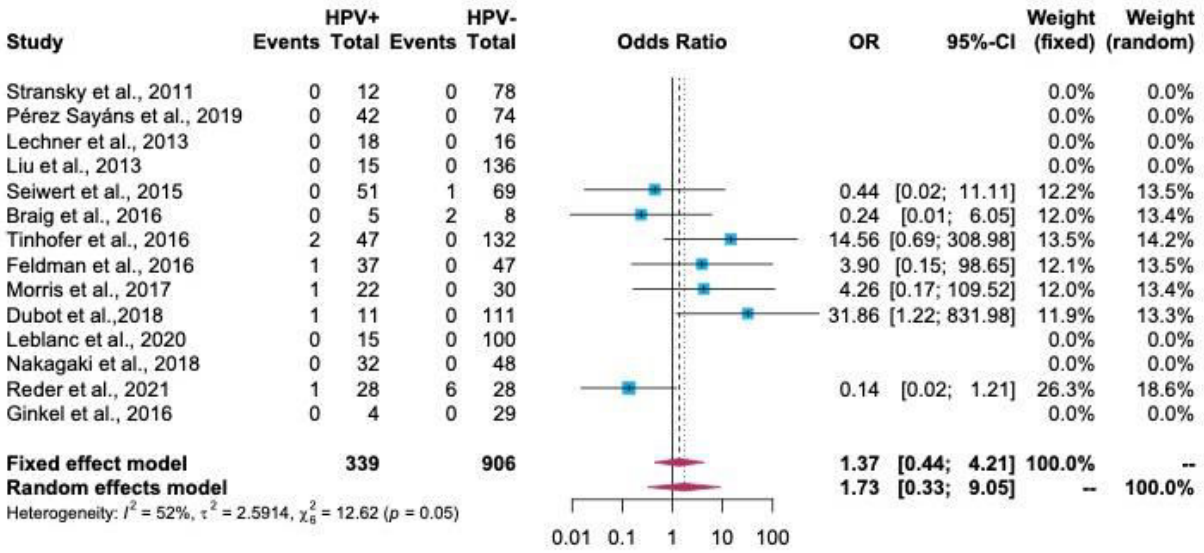


Figure S5: Association between RAS Mutations and HPV status

An odds ratio analysis of the association between human papillomavirus infection status and HRAS or NRAS mutations; no statistically significant correlations were found.

HRAS



NRAS